

can we see what an llm is thinking?

we can record every weight, every activation, and every multiplication. so why does the model still feel like a black box?

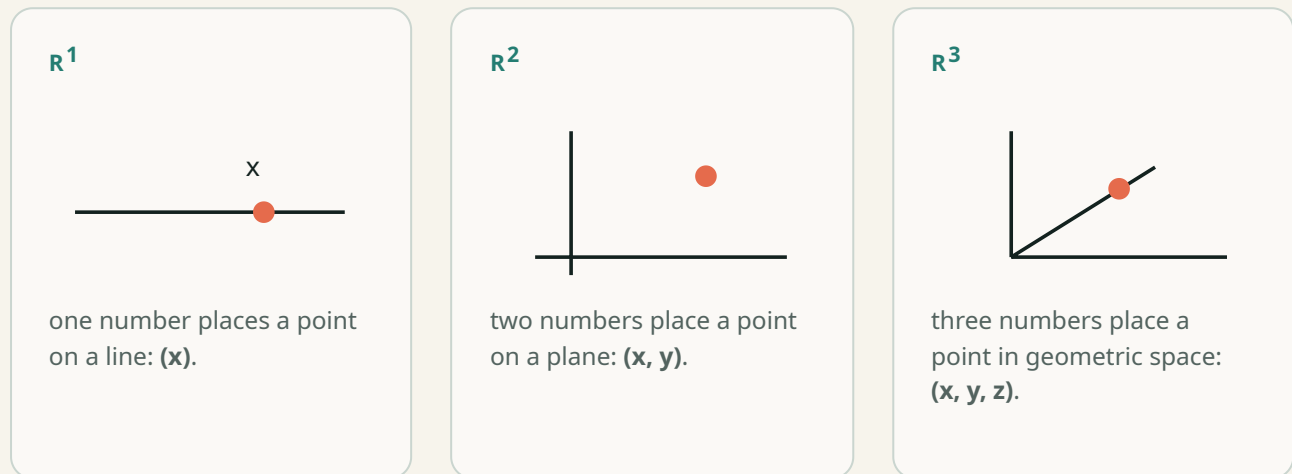


the black box is not hidden. it is illegible.

A step-by-step walk from R^1 to R^{4096} , then into the moving internal state of a transformer.

$\mathbb{R}^{4096}$ is ordinary space with an unreasonable number of directions

the notation looks intimidating, but the idea is familiar. $\mathbb{R}$ means the real numbers, the continuous number line. The superscript tells you how many coordinates are needed to locate one point.



nothing magical happens after three dimensions. We lose the ability to draw the whole space, but the arithmetic keeps working. In $\mathbb{R}^4$, a point has four coordinates. In $\mathbb{R}^{4096}$, it has 4,096.

$$\mathbf{v} = [0.17, -1.02, 0.44, \dots, 0.08] \in \mathbb{R}^{4096}$$

a 4,096-dimensional vector is just a list of 4,096 numbers.

we cannot picture every axis at once, but we can still measure distance, angle, direction, and change.

The examples later use 4,096 and 8,192 because those are real hidden dimensions used by LLaMA model sizes, not universal constants for every LLM.¹

a token does not carry a tiny dictionary definition

before a language model can process text, its tokenizer splits the text into tokens. A token might be a whole word, part of a word, punctuation, or a byte-like fragment. Each token begins with an **embedding**, a learned vector.

$$\text{token} \rightarrow [x_1, x_2, x_3, \dots, x_{4096}]$$

it is tempting to ask which coordinate means “animal,” which means “brown,” and which means “subject of the sentence.” That picture is usually wrong. Useful properties are represented by patterns spread across many coordinates, and the same coordinate can participate in many patterns.

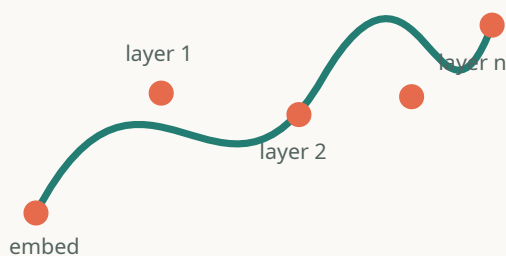
SAME TOKEN, DIFFERENT CONTEXT

bank beside “loan” enters one pattern.

bank beside “river” enters another.

The starting token identity is only the first move. Attention lets surrounding tokens reshape the representation.

ONE POINT BECOMES A PATH



At each layer, the token's hidden state is updated. The vector is a moving point, not a fixed label.

the useful intuition

think of the vector as a location in a giant semantic workspace. Nearby directions can correspond to patterns the model has learned to use, but there is no guarantee those patterns line up neatly with words humans would choose.

do not read the vector coordinate by coordinate. Read it as a whole geometry. Meaning lives in directions, relationships, and transformations.

The original Transformer describes stacked attention and feed-forward layers operating on fixed-width vector representations.²

the model repeats simple operations at enormous scale

a transformer layer is built from familiar mathematical parts: matrix multiplication, weighted sums, normalization, nonlinear functions, and residual updates. None is mysterious alone. The complexity comes from repeating them across many tokens, heads, layers, and parameters.

$$h_{l+1} = h_l + \text{attention}(h_l) + \text{mlp}(h_l)$$

This is a conceptual sketch, not the exact equation for every architecture.

weights

the learned numbers stored in the model. They define how strongly one direction influences another. During ordinary inference, they stay fixed.

activations

the temporary numbers produced when a specific prompt passes through those weights. They change with the input and with every layer.

ONE CONCRETE FAMILY

LLaMA 7B	4,096 dimensions 32 layers
LLaMA 65B	8,192 dimensions 80 layers

These published examples make the scale tangible. Other model families choose other widths and depths.¹

billions of parameters are not billions of little sentences stored in drawers. They are coefficients inside a distributed computation. For each new token, the network uses them to transform a large field of activations, layer after layer, before producing scores for possible next tokens.

the algorithm is inspectable. the emergent organization is not yet readable.

weights stay put. activations flash into existence.

the model's durable object is the network of weights. the “thought” you want to inspect is a temporary activation pattern created by one prompt, at one moment, on one forward pass.

1. INPUT

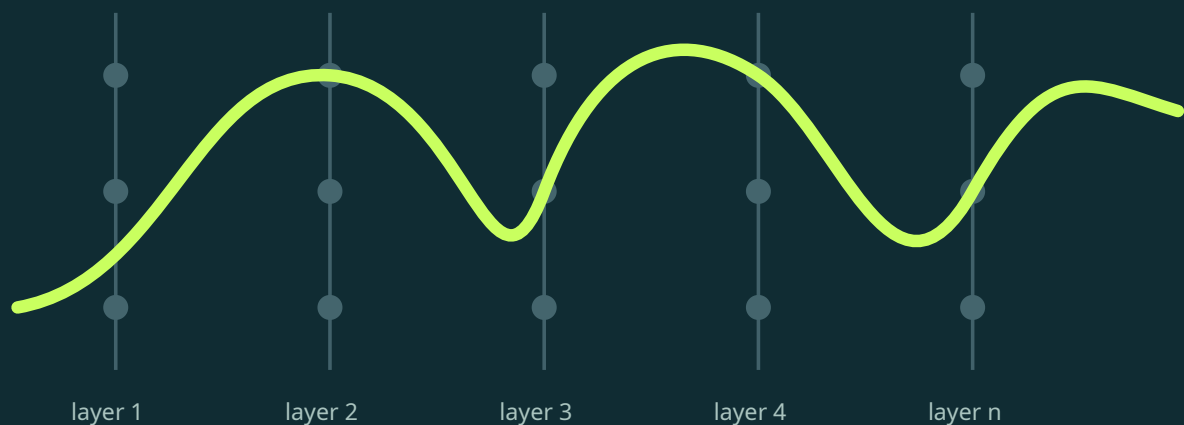
tokens enter with embeddings and positional information.

2. TRANSFORMATION

attention and feed-forward blocks recompute the hidden state.

3. OUTPUT

the final state becomes next-token scores, then one token is selected.



there is no permanent shelf labeled “what the model believes right now.”

Capture the activations and you get a numerical snapshot. Run a different prompt, or continue by one token, and the pattern changes.

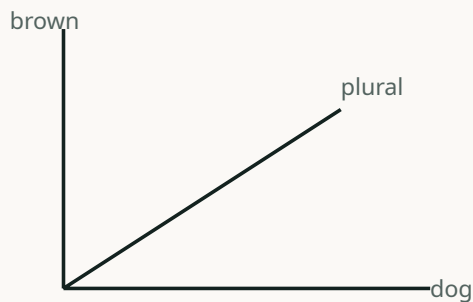
“milliseconds” is the right mental model for many deployed inference systems, but wall-clock timing depends on hardware, model size, context length, and serving setup.

superposition lets many concepts share the same space

suppose a model needs useful features for animals, colors, syntax, places, emotions, programming patterns, and millions of subtler distinctions. Giving every feature its own dedicated coordinate would be wonderfully readable... and terribly wasteful.

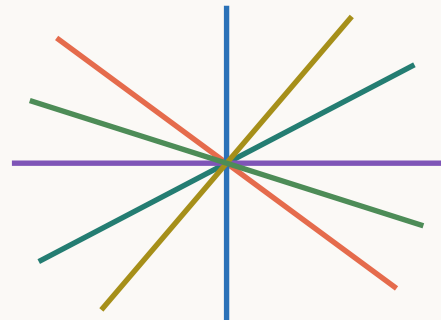
neural networks appear able to pack more features into a representation than there are clean, independent dimensions. A feature can be encoded as a **direction**, and many such directions can overlap in the same space. This is called **superposition**.³

THE TIDY FANTASY



one axis per concept. Easy to label, but too neat to describe how large models actually organize everything.

THE PACKED REALITY



many useful directions reuse the same coordinates. A single neuron can participate in several features.

why “find the dog neuron” usually fails

the concept of a dog may be distributed across a pattern involving many neurons. Meanwhile, one of those neurons may also contribute to fur, pets, loyalty, or sentence structure depending on context. Anthropic summarizes the problem directly: concepts can span many neurons, and neurons can represent many concepts.⁴

the coordinates are visible, but the basis is not privileged. Human concepts do not arrive pre-aligned with the model's axes.

nonlinear mixing makes the path hard to reverse

if every layer performed only a linear transformation, the whole stack could be collapsed into one big matrix. Transformers do more. Attention routes information between token positions, and feed-forward blocks apply nonlinear gates. A feature can be amplified in one context, suppressed in another, or combined with several others.

matrix multiplication

rotates, stretches, projects, and mixes directions in vector space.

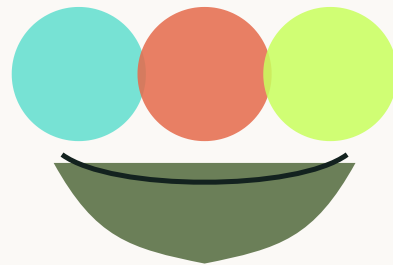
attention

moves information between token positions according to context-dependent weights.

nonlinearity

changes what is active based on thresholds and gates, so the result is not a simple sum you can unwind.

THE PAINT ANALOGY



once three colors are mixed, knowing the final shade does not tell you the exact motion of every drop.

now repeat the process across dozens of layers. A clean story such as “this neuron recognized a dog, so that neuron chose the word bark” is usually too simple. The causal chain is a web of interacting contributions.

tracing a decision backward is less like reading code and more like unmixing paint.

the program itself is explicit. We know the equations. What we do not have is a compact translation from billions of numerical interactions into a faithful human explanation.

interpretability finds real mechanisms, not a complete transcript

the field is not starting from zero. Researchers can record activations, intervene on them, train sparse dictionaries that recover more interpretable features, and trace local circuits that influence an output.

feature dictionaries

Methods such as sparse autoencoders look for recurring activation patterns that behave more like human-readable features than individual neurons do.

causal interventions

Researchers can suppress, amplify, or inject a feature and observe whether the model's behavior changes in the predicted way.

circuit tracing

Attribution graphs can approximate how features pass influence through a model for a particular prompt and output.

behavioral experiments

Carefully designed prompts reveal capabilities and failure modes, though behavior alone does not expose the underlying mechanism.

these tools have produced striking case studies. Anthropic reported tracing mechanisms involved in planning rhymes, multilingual concept sharing, arithmetic, and explanations that do not faithfully reflect the model's internal route.⁵

but a local explanation is not the same as a complete one. Interpretability systems often replace part of the original computation with an easier-to-read approximation. They may recover some active features while missing others. The graph for one prompt does not automatically explain every prompt.

the honest status: we have microscopes, probes, and partial maps. We do not yet have a full source-code-level explanation of what a frontier LLM computes.

what would “solved” look like?

- Faithful explanations that preserve the model's actual causal route.
- Coverage broad enough to audit unfamiliar prompts, not only curated examples.
- Tools that scale with model size without creating a second black box.
- Predictions that survive intervention, replication, and adversarial testing.

we can observe everything and understand almost nothing

that sentence is deliberately sharp, but it captures the gap. The obstacle is not secret code, quantum physics, or a hidden soul inside the machine.

SCALE

billions of learned parameters and vast numbers of intermediate calculations.

DISTRIBUTION

concepts spread across patterns instead of living in one readable location.

DYNAMICS

the relevant internal state is rebuilt for each input and changes token by token.

the mystery is scale and distributed representation... not magic.

machine interpretability is genuinely unsolved. That does not mean it is impossible. It means we are still inventing the language, instruments, and tests needed to translate a moving geometry into a trustworthy explanation.

FIVE IDEAS TO KEEP

1. R^{4096} is a list of 4,096 coordinates, not a mystical realm.
2. A token's vector changes as context moves through the layers.
3. Weights persist; activations are temporary and input-specific.
4. Superposition and nonlinear mixing make direct reading unreliable.
5. Interpretability has real tools and results, but no complete decoder ring.

SOURCES

¹ Touvron et al., LLaMA: Open and Efficient Foundation Language Models, Table 2.

² Vaswani et al., Attention Is All You Need.

³ Elhage et al., Toy Models of Superposition.

⁴ Anthropic, Mapping the mind of a large language model.

⁵ Anthropic, Tracing the thoughts of a large language model.